Facultad de Defensa Nacional

Carrera Lic. En Ciberdefensa

Análisis Matemático I

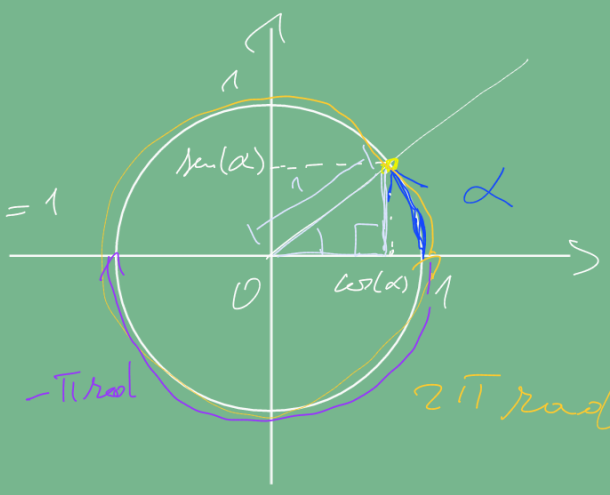
Profesor a cargo: Lic. Nelson Monzón

Profesora adjunta: Prof. Maela Lombardi

Año: 2025

Carga horaria: 64

SISTEMA CIRCULAR



$\sin^2(\alpha) + \cos^2(\alpha) = 1$

$90^\circ = \frac{\pi}{2}$ rad

$45^\circ = \frac{\pi}{4}$

$120^\circ = \cancel{120^\circ}^{2} \cdot \frac{\pi \text{ rad}}{\cancel{180^\circ}_{3}} = \frac{2}{3}\pi$ rad

$\frac{\pi}{6}$ rad $= \frac{\pi}{6}$ rad $\cdot \frac{180^\circ}{\pi \text{ rad}} = 30^\circ$ OJO

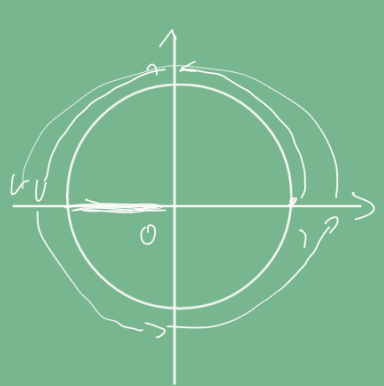
1 rad $= 1$ rad $\cdot \frac{180^\circ}{\pi \text{ rad}} = 57,29\ldots^\circ \cong 57^\circ 17' 44,81''$

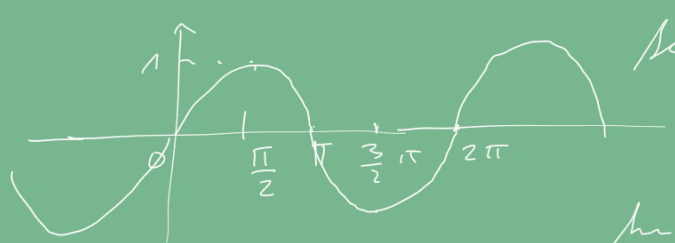
MODE + DEG → GRADOS SEXAGESIMALES

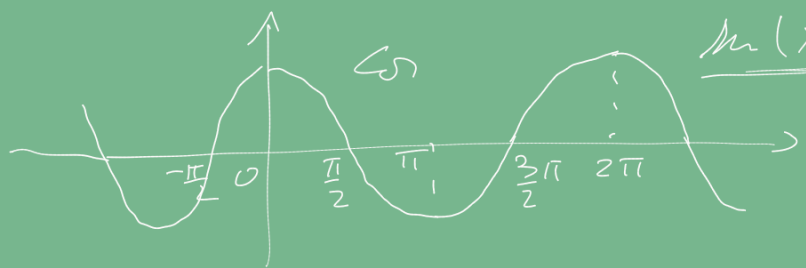
(o SHIFT) RAD → RADIANES.

$tg(x) = \frac{\sin(x)}{\cos(x)}$, $\operatorname{cosec}(x) = \frac{1}{\sin(x)}$

$\sec(x) = \frac{1}{\cos(x)}$, $\operatorname{cotg}(x) = \frac{\cos(x)}{\sin(x)}$







$\sin(x + \frac{\pi}{2}) = \cos(x)$

$\sin(x) = \cos(x - \frac{\pi}{2}) = \cos(\frac{\pi}{2} - x)$

PROPIEDADES:

1) $\operatorname{sen}^2(x) + \cos^2(x) = 1$, $\forall x \in \mathbb{R}$

2) $\operatorname{sen}(-x) = -\operatorname{sen}(x)$; $\cos(-x) = -\cos(x)$, $\forall x \in \mathbb{R}$.

3) $\operatorname{sen}(x \pm y) = \operatorname{sen}(x)\cos(y) \pm \cos(x)\operatorname{sen}(y)$

4) $\operatorname{sen}(2x) = 2\operatorname{sen}(x)\cos(x)$

5) $\cos(x \pm y) = \cos(x)\cos(y) \mp \operatorname{sen}(x)\operatorname{sen}(y)$

6) $\cos(2x) = \cos^2(x) - \operatorname{sen}^2(x)$.

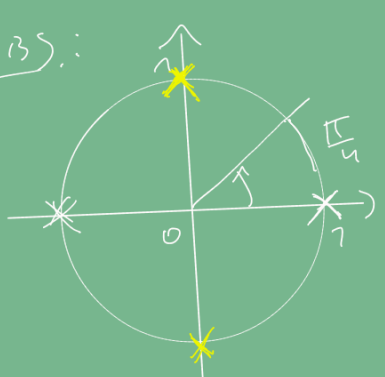
OBS:

$$\cos^2(x) + \operatorname{sen}^2(x) = 1$$
$$+\ \cos^2(x) - \operatorname{sen}^2(x) = \cos(2x)$$
$$2\cos^2(x) = 1 + \cos(2x)$$
$$\boxed{\cos^2(x) = \frac{1 + \cos(2x)}{2}}$$

E.S:

$$\boxed{\operatorname{sen}^2(x) = \frac{1 - \cos(2x)}{2}}$$

OBS:



- $\operatorname{sen}(x) = 0 \iff x = k\pi$, $k \in \mathbb{Z}$.
- $\cos(k\pi) = (-1)^k$, $k \in \mathbb{Z}$.
- $\cos(x) = 0 \iff x = \frac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$
- $\operatorname{sen}\left(\frac{\pi}{2} + k\pi\right) = (-1)^k$, $k \in \mathbb{Z}$

$\cos\left(\frac{\pi}{4}\right) = \operatorname{sen}\left(\frac{\pi}{4}\right) = a$,

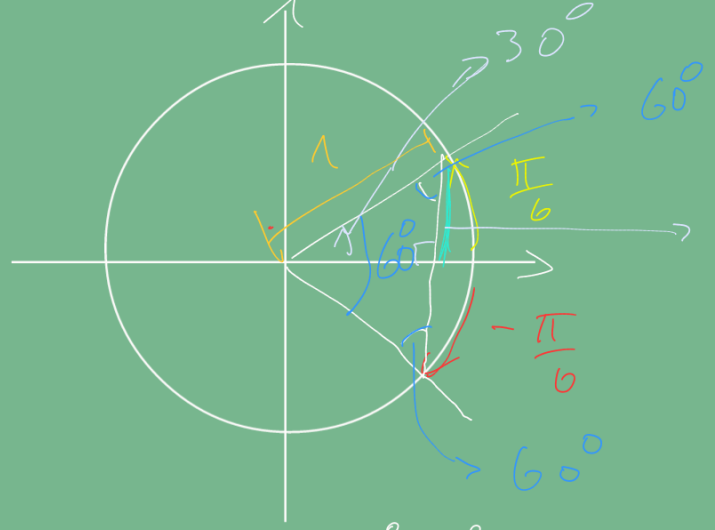
$a^2 + a^2 = 1$

$a^2 = \frac{1}{2}$

$a = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

$\therefore \cos\left(\frac{\pi}{4}\right) = \operatorname{sen}\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$

OBS:



$\frac{1}{2} \Rightarrow \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$

TABLA:

α / $f(\alpha)$	$0°$	$30°$	$45°$	$60°$	$90°$
	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin(\alpha)$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos(\alpha)$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$tg(\alpha)$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$\nexists$

$\cos^2\left(\frac{\pi}{6}\right) + \sin^2\left(\frac{\pi}{6}\right) = 1$

$\cos^2\left(\frac{\pi}{6}\right) + \frac{1}{4} = 1$

$\cos^2\left(\frac{\pi}{6}\right) = \frac{3}{4}$

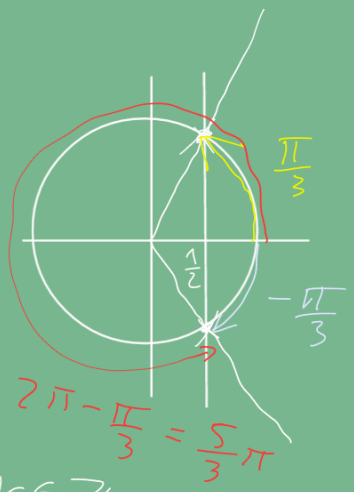
$\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$

$\frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{2}$

Ej.: $2\cos^2(x) - \cos(x) = 0$

$\cos(x) \cdot (2\cos(x) - 1) = 0$

$\cos(x) = 0$ o $\cos(x) = \frac{1}{2}$



$\downarrow$

$x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$

$\rightarrow x = \pm\frac{\pi}{3} + 2k\pi, k \in \mathbb{Z}$

$\therefore S = \left\{\frac{\pi}{2} + k\pi : k \in \mathbb{Z}\right\} \cup \left\{\pm\frac{\pi}{3} + 2k\pi : k \in \mathbb{Z}\right\}$

Ej.: • $\cos(x) = 1{,}02 \longrightarrow S = \emptyset$

• $\sin(x) = -3 \longrightarrow S = \emptyset$

$\left.\begin{array}{l} \operatorname{sen}: \mathbb{R} \longrightarrow [-1,1] \\ \cos: \mathbb{R} \longrightarrow [-1,1] \end{array}\right)$ SOBREYECTIVAS

Ej: $\operatorname{sen}^2(x) - 5\operatorname{sen}(x) + 6 = 0$

$t = \operatorname{sen}(x) \Rightarrow \quad t^2 - 5t + 6 = 0$

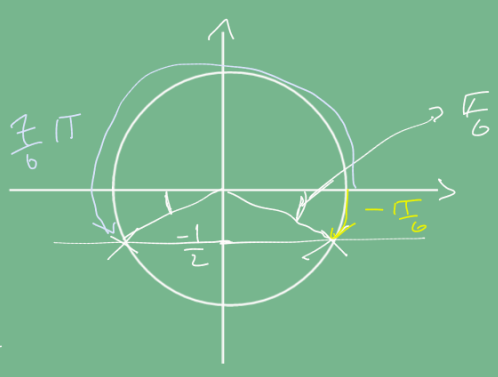
$(t-2).(t-3) = 0$

$t = 2 \quad \text{o} \quad t = 3$

$\operatorname{sen}(x) = 2 \quad \text{o} \quad \operatorname{sen}(x) = 3$ FALSO

$\therefore \boxed{S = \emptyset}$

Ej: $\operatorname{sen}(5x) = -\frac{1}{2}$



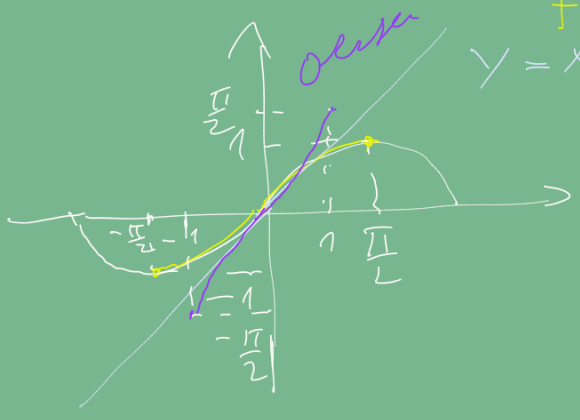
$5x = -\frac{\pi}{6} + 2k\pi, \; k \in \mathbb{Z}$

o

$5x = \frac{7}{6}\pi + 2m\pi, \; m \in \mathbb{Z}$

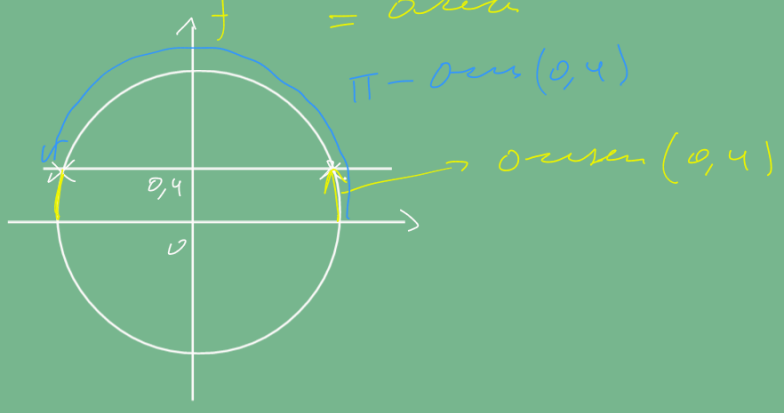
$$S = \left\{ -\frac{\pi}{30} + \frac{2k\pi}{5} : k \in \mathbb{Z} \right\} \cup \left\{ -\frac{7}{30}\pi + \frac{2m\pi}{5} : m \in \mathbb{Z} \right\}$$

Ej: $\operatorname{sen}(x) = 0{,}4$

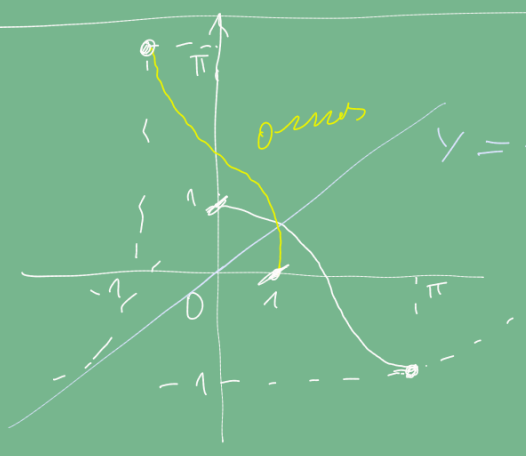


$f: [-\frac{\pi}{2}, \frac{\pi}{2}] \longrightarrow [-1,1]$

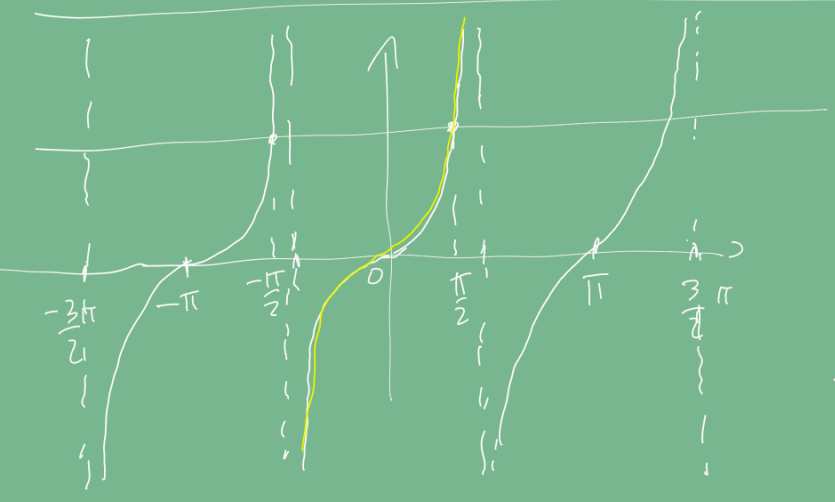
$f^{-1}: [-1,1] \longrightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$



$$S = \{\arccos(0,4) + 2k\pi : k \in \mathbb{Z}\} \cup \{\pi - \arccos(0,4) + 2m\pi : m \in \mathbb{Z}\}$$



$$\arccos : [-1,1] \longrightarrow [0,\pi]$$

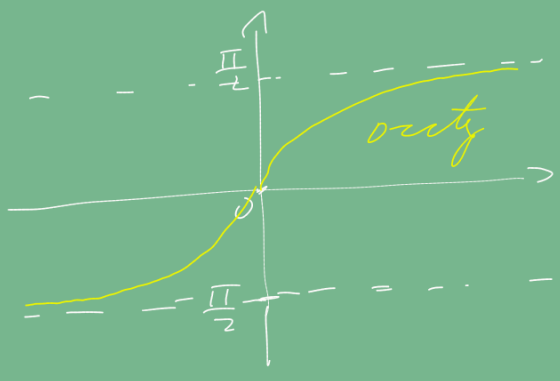


$$tg(x) = \frac{\sin(x)}{\cos(x)}$$

$$tg(x+k\pi) = tg(x) \rightarrow (k \in \mathbb{Z})$$
$$\hookrightarrow (x \neq \tfrac{\pi}{2} + m\pi, m \in \mathbb{Z})$$

$$tg(\alpha) = tg(\beta) \Leftrightarrow \exists k \in \mathbb{Z} / \alpha - \beta = k\pi$$

$$\operatorname{arctg} : \mathbb{R} \longrightarrow (-\tfrac{\pi}{2}, \tfrac{\pi}{2})$$



Ej: $\ln(2x+y) = \ln(2x) \cdot \ln(y)$

$x = y = 1 \Rightarrow \ln(3) = \ln(2) \cdot \underbrace{\ln(1)}_{=0} \Rightarrow \ln(3) = 0$ FALSO

$\log_e(3) = 0$

FALSO

$3 = e^0 = 1$

FALSO